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PERFORMANCE SPECIFICATION
DETECTOR-DEWAR, DT-591B/UA

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers one type of common module, Detector-Dewar, DT-591B/UA, herein referred to as the Detector-Dewar.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.2.1 Handbooks. The following handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and, therefore, are not mandatory.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - Standard General Requirements for Electronic Equipment.

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2). Drawings listed below are to be used to ensure accurate interface of the item produced under this performance specification with existing hardware.

PURCHASE DESCRIPTION

DRAWINGS

USA CECOM

80063-A3172178 - Detector-Dewar DT-591B/UA
80063 SM-C-969732 Bellows

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Standard Procedures for Testing Infrared Detectors and Their Performance, R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering.

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(Copies of the above non-government publication should be obtained from the procuring activity or as directed by the contracting officer).

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z 1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 53202).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract (see 6.2), the contractor shall furnish first article units. A sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The contractor shall select materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria for the class, grade or type part shall ensure that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Detector-Dewar, DT-591B/UA, is an opto-electronic device sensitive to radiation in the 7.5 to 12.0 micrometer spectral region. The Detector-Dewar uses a supply of photoconductive detectors operated from a power supply of 3.0, ± 0.2 volts direct current (Vdc). Bias resistors shall be an integral part of the Detector-Dewar.

3.3.1 Design. The Detector-Dewar shall meet the form, fit, function and interface requirements of this specification.

3.3.2 Weight. Maximum weight of the Detector-Dewar shall be 0.5 pounds including bias resistors.

3.3.3 Dimensions. Dimensions and dimensional tolerances of the Detector-Dewar shall be in accordance with 80063-A3172178 to esnure proper physical interface with the existing hardware.

3.4 Components. The Detector-Dewar shall consist of a detector array and a resistor bias pack. The detector array shall rotate 360° with respect to the Dewar mounting flange. Two or more reusable, electrically activated getters shall be an integral part of the Detector-Dewar for

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maintenance of the vacuum. Each getter shall withstand at least seven (7) firings in operational use. Activation current (amperes) and time (minutes) shall be provided on the label as shown in View K on Drawing 80063-A3172178. Data shall be marked in black characters on a light background in accordance with Note 27 of Drawing 80063-A3172178.

3.5 Performance characteristics.

3.5.1 Detector-Dewar defects. The Detector-Dewar shall operate at 80°, +5/-20°K. The Detector-Dewar shall contain no more than two (2) defective elements, as defined in 3.12.2. The center twelve (12) elements shall be defect-free. Defects shall be separated from one another by at least four (4) nondefective elements. Elements which are open (dead) or have a D*BB or less than 7.0×10^{9} shall be electrically strapped to an adjacent element with common ground (perform 4.5.1 for acceptance).

3.5.2 Time constant. The time constant is the time for the element response to reach 63% of its steady state value when irradiated by a 7.5-12 micrometer source producing a signal to noise ratio of <100:1. The time constant of each Detector-Dewar element shall be less than 2 microseconds (perform 4.4.2 for acceptance).

3.5.3 Sensitive area. The physical area shall be 0.0960 by 0.0144 inches or greater. Perform 4.4.3 for acceptance.

3.5.4 Crosstalk. The crosstalk signal level between any 2 elements shall be less than -26 dB as determined by measuring 4 elements. Perform 4.4.4 for acceptance.

3.5.5 Blackbody detectivity. The blackbody detectivity of all nondefective elements shall not be less than 10.5×10^{9} cm $Hz^{½}$ w^{-1}, except for channels 85-96, which shall not be less than 11.5×10^{9} cm $Hz^{½}$ w^{-1} when the Detector-Dewar is cooled with LN_2 to 77 ±3°K (perform 4.4.5 for acceptance). Noise measurements shall be made on each element of the array under the following conditions: The Detector-Dewar cooled to 77 ±3°K with LN_2, 295, ±5°K background temperature and equivalent noise bandwidth of 9 KHz ±200 Hz with low frequency 3 dB point (high pass) no greater than 80 Hz. Using the results of these measurements, the results of 500°K blackbody responsivity measurements and the nominal detector element size (see 3.12.7), the 500°K blackbody detectivity for each element of the array shall be calculated.

3.5.5.1 Noise. The detector noise of each element shall be at least 2.5×10^{-9} volts rms $Hz^{1/2}$.

3.5.6 Detector Resistance. The detector resistance of each element shall be not less than 20 ohms and not greater than 150 ohms.

3.5.7 Responsivity. The average 500°K blackbody responsivity of the acceptable elements shall not be less than 10.3×10^{3} volts/watt, nor greater than 20×10^{3} volts/watt, when the Detector-Dewar is cooled with LN_2 to 77°, ±3°K. The blackbody responsivity of each element shall not vary by more than ±20 percent of the mean value of the elements which are not defective. Nominal

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detector element size shall be used in responsivity calculations and blackbody temperature shall be 500°, ±5°K (see 3.12.7).

3.5.8 Temperature sensors. At least two calibrated 2N2222 temperature sensor chips shall be installed at or near the detector array location in the Dewar to provide an electrical signal which is an analog of temperature. The temperature sensor shall have a junction voltage of 1.060, ±0.015 volts with a 1.00, ±0.01 milliamperes (mA) forward current when the Detector-Dewar is operated at 77°, ±3°K. The voltage shall be recorded on the appropriate label on the Detector. The 77°K calibration point shall be determined by cooling the inner Dewar with LN_2. A 1.00, ±0.01 mA bias shall be applied to the sensor diode and the voltage shall be monitored and recorded at the 77°K calibration point.

3.5.9 Spectral Response. Each element of the Detector-Dewar shall have special sensitivity, defined as the 50 percent relative response points, of 7.7, ±0.3 to 11.7, ±0.5 micrometers. The spectral response of the sixth element from each of the array or adjacent element on the high end and the higher number of the two center elements shall be tested per “Standard Procedures for Testing Infrared Detectors and Describing Their Performance”, by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. If these elements are defective an adjacent element may be substituted.

3.5.10 Detector bias. The average bias current of the array shall not exceed 3.0 mA per element, and no one element shall take a bias current of more than 4.0 mA. The detector bias power shall be tested as per 3.5.9, above. The total bias power dissipated by the array shall not exceed 30 milliwatts (mW). The detector element shall be biased with current flow in the same direction. The detector bias pack shall be operated at 3.0, ±0.2 Vdc.

3.5.11 Detector-Dewar warm and cold shields. The incorporation of an internal cold shield is optional. If incorporated, the Detector-Dewar shall meet the following Field Of View (FOV) requirements: The minimum half angle FOV from the center of the end elements of the array in the elevating direction shall be 14 degrees. In the direction perpendicular to the array’s axis, the cold shield(s) shall allow for each element, no vignetting of any ray to any point on the detector surface within an angle of 17 degrees. This shall be measured from and on either side of a line perpendicular to the array’s axis and parallel to the optical axis. During in-process inspection, the warm and cold shields shall be measured to ensure conformance with the requirement. Alignment of the cold shield relative to detector elements shall be checked 100 percent during in-process inspection.

3.5.12 Heat load. The detector heat load shall not exceed 120 mW when all detector elements are operating at their final bias current and the Detector-Dewar is cooled to 80°, ±5°K.

3.5.13 Vacuum integrity. The Dewar shall comply with heat load requirements of 3.5.12 for a period of at least one (1) year.

3.5.14 Microphonics. When cooled to a temperature of 60-80°K, with full operating bias applied in accordance with paragraph 3.5.10, and vibrated in accordance with paragraph 3.5.14.1,

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all elements of the Detector-Dewar shall have no noise peaks above the white noise reference value greater than 8 dB from 300 to 10,000 Hz, and 12 dB from 10,001 to 20,000 Hz. using a 191 Hz, Bandwith, Flat Top window. Vibration excitation shall be measured by an accelerometer mounted on the mounting plate of the Detector-Dewar. The white noise reference level may be determined by averaging the noise in each channel from 12,000 to 20,000 Hz with vibration applied. Noise spectra shall be measured using a bandwidth between 50 Hz and 365 Hz.

3.5.14.1 Microphonics Vibration. The Detector-Dewar shall be coupled to a mechanical cryogenic refrigerator that produces random vibrational excitation of the Detector-Dewar in three mutually perpendicular axes. The frequency distribution of the vibrations shall be in a range from 0 to 20,000 Hz. The rms sum of the three axis vibrations shall be at least 1.0 g rms. Each axis shall produce a minimum vibrational excitation of 0.4 g rms. In the frequency range of 2,000 to 20,000 Hz, the rms sum shall be at least 0.8 g rms.

3.5.15 Apparent thermal mass (cooldown). When mated to a cryogenic cooler capable of cooling a test mass of 1.25 grams (250 joules) of aluminum (Figure 3) from ambient (296° K) to 80° K in not less than 5 nor greater than 6 minutes, the Detector-Dewar cooldown time from ambient temperature to 80° K shall not exceed the time achieved with the test mass. This measurement shall be made using a highly polished metal Detector-Dewar without any radiation shielding. This test Detector-Dewar and test mass shall consist of the following: A high vacuum of less than 10 microtorr, and a 1.25 gram aluminum test mass, (see B2-2303020104 paragraph 4.2.1). The time to cool the thermal mass shall be recorded. The voltage output of the temperature sensor diodes in the Detector-Dewar under test shall be measured at 77 ±1° K and at ambient temperature. The voltage output at 80° K shall then be interpolated. The cooler shall then be coupled to the Detector-Dewar using thermal grease and bellows. Operating bias power shall be applied and the time for the temperature of the diode to reach 80° K shall be recorded.

3.6 Environmental conditions.

3.6.1 Temperature shock. The Detector-Dewar shall not be damaged (see 3.12.6) by sudden changes in temperature between -54° C and +71° C (see 4.5.1).

3.6.2 High temperature. The Detector-Dewar shall not be damaged by storage or operation to 71°C (see 3.5.2).

3.6.3 Low temperature. The Detector-Dewar shall not be damaged by operation to -54° C or storage to -62°C.

3.6.4 Shock. The Detector-Dewar shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.6.5 Vibration. The Detector-Dewar shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in Figure 1.

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3.7 Reliability. The Detector-Dewar shall have a specified mean time between failure (MTBF) of 27,600 hours.

3.8 Treatment and painting. Unless otherwise specified, the Detector-Dewar shall have an external surface of a composition or surface treatment which will prevent rust, corrosion and fungus growth in the full range of environmental requirements. External color and surface shine shall reduce light reflection and reduce recognition. Surface treatment adherence must be assured through the full range of environmental and use factors.

3.9 Preconditioning. Each Detector-Dewar shall be subjected to vibration and temperature cycling prior to Group A testing as specified in Figure 2.

3.10 Nameplates and product marking. The Detector-Dewar module, parts, components, subassemblies and assemblies thereof shall be marked for identification and reference designation markings in accordance with View J on 80063-A3172178. Marking shall be positioned as shown on 80063-A3172178.

3.11 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires or integrated circuits shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4 or 5, mandatory for this specification.

3.12.1 Element. An element is defined as one channel in the array of elements in the Detector-Dewar.

3.12.2 Defective element. A defective element shall be defined as an element which fails to meet any one of the requirements of 3.5.5, 3.5.5.1, 3.5.6, 3.5.7 and 3.5.14.

3.12.3 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, shall be considered as a single defect. Elements whose D* is less than 7.0 x 109 $cm.Hz^{1/2}w^{-1}$ shall be opened prior to strapping.

3.12.4 Array axis. The array axis is defined as a line on the surface of the array passing through the center of the projected geometrical areas of each element.

3.12.5 Optical axis. The Detector-Dewar's optical axis is defined as a line perpendicular to the detector mounting plate that intersects the array axis at the midpoint of the array.

NOTE: One-half of the active detector elements are located on each side of the optical axis.

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3.12.6 Damage. Breakage, loosening, shifting, evidence of corrosion or failure of any finish, hardware, connection or component; and any degradation of Detector-Dewar performance to values less than specified herein.

3.12.7 Nominal detector element size. Defined as 0.0016 by 0.0024 inch.

3.12.8 Variation of responsitivity with temperature. When coupled to a cryogenic refrigerator the temperature of the array may reach 55K under these conditions and the responsivity (paragraph 3.5.7) may decrease to as low as 5.6×10^3 volts/watt.

3.12.9 Microphonic test caution. The microphonic requirements and test specified herein are designed to be representative of an operational environment and combat scenario for detectors mounted in a well designed system using current technology. If the system vibrational levels are increased above those specified, the microphonic noise may increase.

3.12.10 Detector temperature. Detector-Dewar temperature is defined as that indicated by the temperature sensor diode.

4. VERIFICATION

4.1 Classification of inspections. The inspections shall be classified as follows:

a. First article inspection (4.2).

b. Conformance inspection (4.3).

4.2 First article inspection. When specified in the contract, first article inspection shall be performed by the contractor on a minimum of 5 preconditioned (see 3.9) sample Detectors-Dewar.

4.2.1 Inspections. All completed assemblies shall be examined for conformance to the applicable specifications or drawings.

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TABLE I. Inspections.

Inspection	Requirement paragraph
Dimensions not as specified	3.3.1
Any as specified part or component missing or damaged	3.3.1
Weight not specified	3.3.2
Getters and label not as specified	3.4
Detector-Dewar orientation range not specified	3.4
Sensitive area not as specified	3.5.3
Detector-Dewar warm and cold shields not as specified	3.5.11
Treatment and painting not as specified	3.8
Marking not as specified	3.10
Workmanship not as specified	3.11

4.2.2 Tests. First article inspection shall be conducted upon units which have successfully completed the inspections specified in 4.2.1. Tests may be conducted in any order, except that the microphonics inspections shall be performed after shock and vibration tests have been completed (see 3.6.4 and 3.6.5).

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Verification paragraph
Detector-Dewar defects	3.5.1	4.4.1
Time constant	3.5.2	4.4.2
Crosstalk	3.5.4	4.4.4
Detectivity	3.5.5	4.4.5
Detector resistance	3.5.6	4.4.6
Responsivity	3.5.7	4.4.7
Temperature sensors	3.5.8	4.4.8
Spectral response	3.5.9	4.4.9
Detector bias	3.5.10	4.4.10
Heat load	3.5.12	4.4.12
Vacuum integrity	3.5.13	4.4.13
Microphonics	3.5.14	4.4.14
Apparent thermal mass (Cooldown)	3.5.15	4.4.15
Temperature shock	3.6.1	4.5.1
High temperature	3.6.2	4.5.2
Low temperature	3.6.3	4.5.3.
Shock	3.6.4	4.5.4
Vibration	3.6.5	4.5.5
Reliability	3.7	4.6

4.2.3 Disposition of first article samples. First article sample shall not be considered as part of the procurement quantities (see 6.2).

4.2.4 Inspection procedures. Unless otherwise specified herein, the Detector-Dewar shall be operated at an input voltage of 3.0 ±0.2 Vdc at ambient temperature (see 6.4) when operated at temperature of 80K, ±5°K, and biased to +3.0,±0.2 Vdc.

4.3 Conformance inspection.

4.3.1 Group A Inspection. Group A inspection shall be conducted on all (100 percent) of the Detector-Dewars following the successful inspection of 4.2 and preconditioning of 3.9. Group A tests listed in Table III may be performed in any order except that Detector-Dewar defects (see 3.5.1) shall be run last.

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TABLE III Group A inspection.

Inspection	Requirement paragraph	Verification paragraph
Detector-Dewar defects	3.5.1	4.4.1
Detectivity	3.5.5	4.4.5
Responsivity	3.5.7	4.4.7
Detector resistance	3.5.6	4.4.6
Temperature sensors	3.5.8	4.4.8
Detector bias	3.5.10	4.4.10
Heat load	3.5.12	4.4.12
Microphonics	3.5.14	4.4.14

4.3.2 Group B inspection. Not required.

4.3.3 Group C inspection. Group C inspection shall be conducted on Detector-Dewars selected from units which have passed the inspection in 4.3.1. The performance requirement measurements of 3.5.9, 3.5.13, 3.5.15 and 3.6.5 shall be performed on a sample quantity, selected in accordance with the contract (see 6.2).

4.3.3.1 Group C failures. Action required relative to group C failures shall be as specified the contract or purchase order. (see 6.2.)

4.3.3.2 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to successful completion of testing of Table III.

4.3.4 Group D inspection. Not required.

4.4 Methods of inspection.

4.4.1 Detector-Dewar defects. The detector array shall be checked for defects. Failure to meet requirements of 3.5.1 shall constitute failure of this test.

4.4.2 Time constant. Measurement shall be performed on two (2) elements of the array each separated by at least 20 elements. This time constant is defined as the time required for the element response to reach 63 percent of its steady state value when irradiated by a 7.5 to 12 micrometer source which produces a signal-to-noise ratio not greater than 100:1. This time constant shall be considered equal to the time constant defined by the high frequency rolloff of the noise spectrum of the detector. Any measurement technique consistent with these definitions shall be considered acceptable. Failure to meet requirements of 3.5.2 shall constitute failure of this test.

4.4.3 Sensitive area. The physical area of 0.0960 by 0.0144 inches or greater shall be determined by optical microscope examination IAW drawing A3172178. Measurement shall be made on two (2) elements of the array each separated by at least 20 elements prior to the integration of the Dewar window with the Detector-Dewar. Failure to meet requirements of 3.5.3 shall constitute failure of this test.

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4.4.4 Crosstalk. Measurements shall be made on four (4) elements of the array. The crosstalk shall be measured under the following conditions:

a. An element shall be irradiated by a 7.5 to 12 micrometer source having a signal 20 times greater than the noise level for the element. The image of the source shall fall entirely within the area of the detector element.

b. The signal level of the element adjacent to and on the same ground path of the irradiated element shall be measured. The element under test shall meet the requirements of 3.5.4. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

4.4.5 Blackbody detectivity. Noise measurements shall be made on each element of the array under the following conditions: The detector is cooled to 77K ±3K with LN_2, 295K, $\pm 5^{o}$ K background temperature and equivalent noise bandwidth of 9 kHz ±200 Hz with low frequency 3dB point (high pass) no greater than 80 Hz. Using the results of these measurements, the results of the 500K blackbody responsivity measurements and the nominal detector size (see 3.12.7), the 500K blackbody detectivity for each element of the array shall be calculated. Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.4.6 Detector resistance. The resistance of each element of the array shall be measured. Failure to meet requirements of 3.5.6 shall constitute failure of this test.

4.4.7 Responsivity. Blackbody responsivity of each element of the array may be measured with the Detector-Dewar cooled with LN_2 to 77K ± 3K. Nominal element size (see 3.12.7) shall be used in responsivity calculations and blackbody temperature shall be 500°, ±5K. Failure to meet the requirements of 3.5.7 shall constitute failure of this test.

4.4.8 Temperature sensors. The 77K calibration point may be determined by cooling the inner Detector-Dewar with liquid nitrogen. A 1.00 mA ±0.01 mA bias shall be applied to the sensor diode and the voltage shall be monitored and recorded at the 77K calibration point. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.4.9 Spectral response. The spectral response of the sixth element from each end of the array or adjacent element on the high end and the higher number of the two center elements shall be tested per “Standard Procedures for Testing Infrared Detectors and Describing Their Performance”, by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. If these elements are defective, an adjacent element may be substituted. The spectral sensitivity is defined as the 50 percent relative response points and shall be measured on the assembled Detector-Dewar. Failure to meet requirements of the 3.5.9 shall constitute failure of this test.

4.4.10 Detector bias. The detector bias power shall be tested per “Standard Procedures for Testing Infrared Detectors and Describing Their Performance”, by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. Failure to meet the requirements of 3.5.10 shall constitute failure of this test.

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4.4.11 Dewar warm and cold shields. The warm and cold shields shall be measured during in-process inspection. With this requirement, alignment of the cold shield relative to detector elements shall be checked 100 percent during in-process inspection. Failure to meet the requirements of 3.5.11 shall constitute failure of this test.

4.4.12 Heat load. The heat load of the Detector-Dewar shall be determined with all detectors operating at specified bias. The heat load of the Detector-Dewar shall be determined by partially filling the Detector-Dewar's inner stem with liquid nitrogen and monitoring the rate of loss of the cryogen. The heat load shall be calculated from the average or integrated loss rate of the cryogen for the last 5 minutes of the Detector-Dewar's hold-time. For calculations, cooling produced by the nitrogen phase change only shall be considered as total useful energy. If this test is performed with the detector array unbiased, the results of this test shall be added to the results of the detector bias power test to determine the heat load. Failure to meet the requirements of 3.5.12 shall constitute failure of this test.

4.4.13 Vacuum integrity. Vacuum integrity shall be determined on a basis of heat load capacity. Heat load shall be measured on the unbiased Detector-Dewar. The temperature of the Detector-Dewar shall then be elevated to +71°C for a period of 72 hours. At the completion of the temperature exposure and stabilization at ambient temperature, the heat load shall be measured again. Getters may not be fired during this test. Failure of the Detector-Dewar to meet the requirements of 3.5.12 or a 15 percent increase in heat load between the measurement before and after temperature exposure, shall constitute failure of this test.

4.4.14 Microphonics. Microphonics shall be measured while the Detector is vibrated in accordance with paragraph 4.5.14.1. The Detector-Dewar shall be cooled to 60K-80K with full operating bias applied. Mounting to the cooler shall be made by tightening the mounting screws to a torque of 3.50 inch pounds minimum. Vibration excitation shall be measured by an accelerometer mounted on the mounting plate of the Detector-Dewar. The white noise reference level may be determined by averaging the noise in each channel from 12,000 to 20,000 Hz with vibration applied. Noise spectra shall be measured using a bandwidth between 50 Hz and 365 Hz. Failure of any channel to meet the requirements of 3.5.14 shall constitute a defect as defined in paragraph 3.5.1.

4.4.14.1 Microphonics Vibration. The Detector-Dewar shall be coupled to a mechanical cryogenic refrigerator that produces random vibrational excitation of the Detector-Dewar in three mutually perpendicular axes. The frequency distribution of the vibrations shall be in a range from 0 to 20,000 Hz. The rms sum of the three axis vibrations shall be at least 1.0 g rms. Each axis shall produce a minimum vibrational excitation of 0.4 g rms. In the frequency range of 2,000 to 20,000 Hz, the rms sum shall be at least 0.8 g rms. The mechanical cryogenic refrigerator shall be mated to the Detector-Dewar with a metal bellows SM-C-969732 during the test for this requirement.

4.4.15 Apparent thermal mass. (Cooldown). An HD-1132/UA cryogenic cooler shall be mated to the standard test mass (Figure 3). The capacity of the HD-1132/UA shall be such that the 1.25 gram (250 joules) test mass is cooled from ambient (296K) temperature to 80K ±1.5K in not less than 5 nor greater than 6 minutes. This measurement shall be made using a highly

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polished metal Detector-Dewar without any radiation shielding. This test Detector-Dewar and test mass shall consist of the following: A high vacuum of less than 10 microtorr, and a 1.25 gram aluminum test mass. The time to cool the thermal mass shall be recorded. The voltage output of the temperature sensor diodes in the Detector-Dewar under test shall be measured at 77K ± 1K and at ambient temperature. The voltage output at 80K shall then be interplated. The cooler shall then be coupled to the Detector-Detector using thermal grease and bellows. Operating bias power shall be applied and the time for the temperature of the diode to reach 80K shall be recorded. Failure of the diode to cool within the requirements of 3.5.15 shall constitute failure of this test.

4.5 Environmental tests. Environmental testing methods contained in 6.4 are acceptable methods for determining compliance with Section 3.6, et al. The contractor may elect to use alternative methods and testing procedures. In such cases, the contractor shall be required to show where his own tests and measurements for verifying the environmental conditions are equivalent to the methods cited herein. Where the contractor desires to deviate from these cited methods, he shall be prepared to show equivalency via a test plan which has been approved by the contracting officer. Also, environmental tests shall be accomplished using testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented. The methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to the recommended procedures found herein.

4.5.1 Temperature shock. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.1. A recommended method for verification may be found in 6.4.

4.5.2 High temperature. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.2. A recommended method for verification may be found in 6.4.

4.5.3 Low temperature. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.3. A recommended method for verification may be found in 6.4.

4.5.4 Shock. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.4. A recommended method for verification may be found in 6.4.

4.5.5 Vibration. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.5. A recommended method for verification may be found in 6.4.

4.6 Reliability. A reliability prediction and math model shall be maintained by the contractor which reflect the current design and components of the Detector-Dewar. Failure of the prediction or model to explicitly delineate compliance with this section shall constitute failure of the reliability requirement (see 3.7).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department system Command. Packaging data retrieval is available form the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The Detector-Dewar is intended for use in infrared systems.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required:

(1) Time frame for submission of first article test reports are required.

(2) Time frame for approval of first article test reports are required.

(3) First article reliability testing, if required.

c. Sampling plans for Groups B and C.

d. When first article rough handling tests are not required.

e. Necessary actions by the contractor in the event of a Group C failure (see 4.3.3.1).

f. Deliver all first article samples to the procuring activity's contracting officer.

g. Packaging requirements (see 5.1).

h. The issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2. and 2.3).

6.3 Definitions. (see 3.12).

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6.4 Environmental inspection procedures. The test-methods and procedures of MIL-STD-810 listed below are recommended, but are not mandatory for determining compliance with the environmental requirements of this specification.

Inspection	Spec. Paragraph	MIL-STD-810 Method	MIL-STD-810 Procedure	Remarks
Temperature shock	3.5.1	503.1	I	For Step 2 use -54°C and steps 1 and 4 use +71°C
High temperature	3.6.2	501.1	I	For steps 4 and 4 use +71°C
Low temperature	3.6.3	502.1	I	For storage temperature, step 2, use -62°C and low temperature use -54°C
Shock	3.6.4	516.2	IV, V	Procedure IV use Figure 516.2-1
Vibration (see Figure 1)	3.6.5	514.2	I, VIII	Use Figure 1 in place of MIL-STD-810 curve.

6.5 Subject term (key word listing).

Flir Systems
Opto-electronic
Sensors

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR

(Project 5855-0090)

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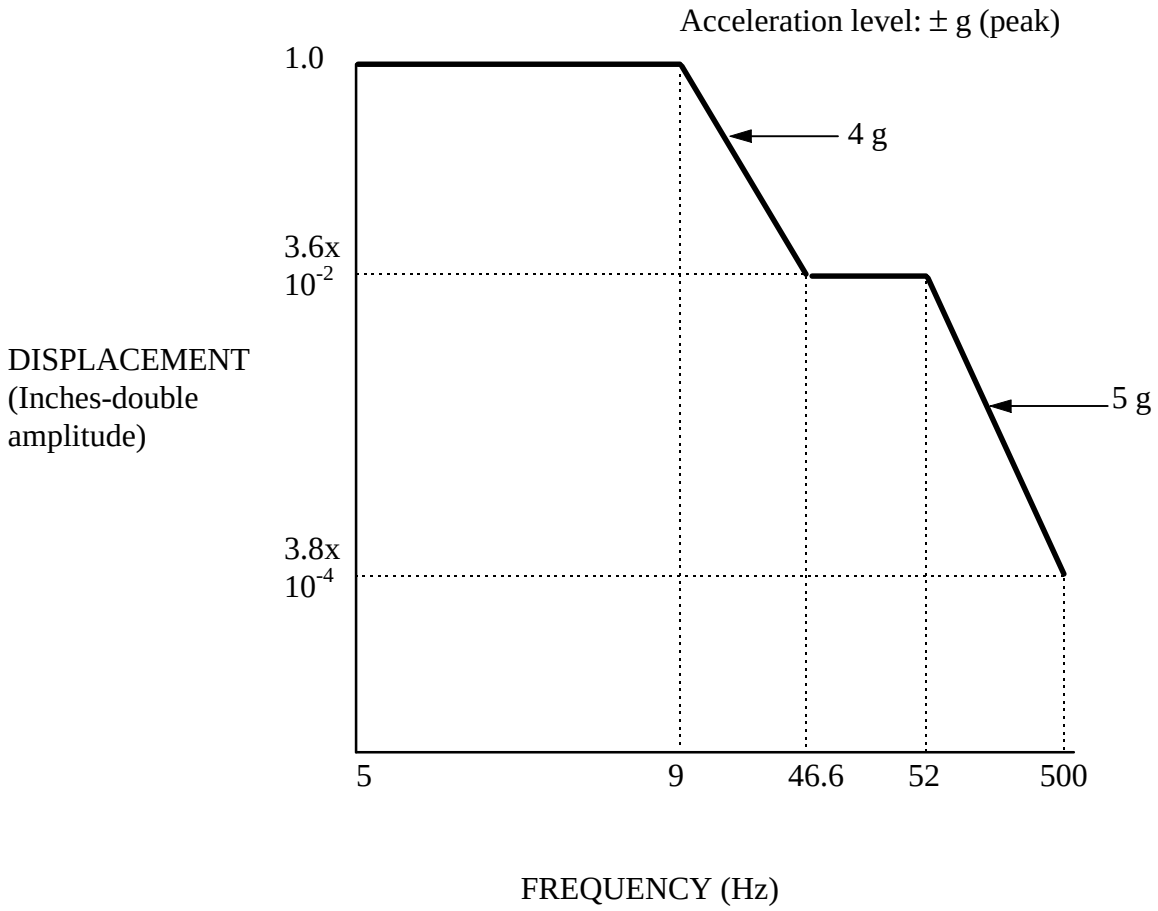
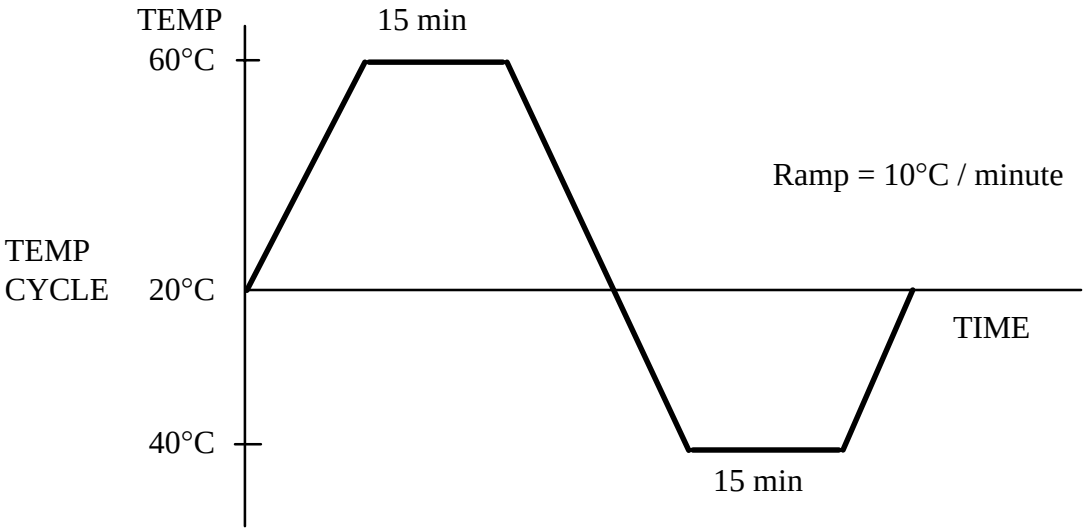
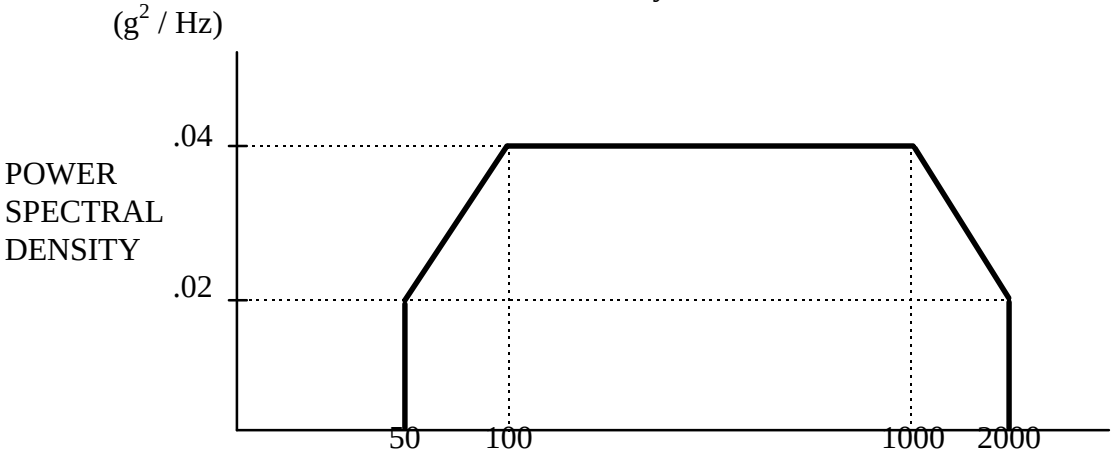


FIGURE 1. Vibration Test Profile.

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Vibration cycle[1] : 15 minutes



Notes:

1. Vibration performed along detector cold well axis.
2. Temperature cycle precedes vibration cycle.

FIGURE 2. Preconditioning Cycle.

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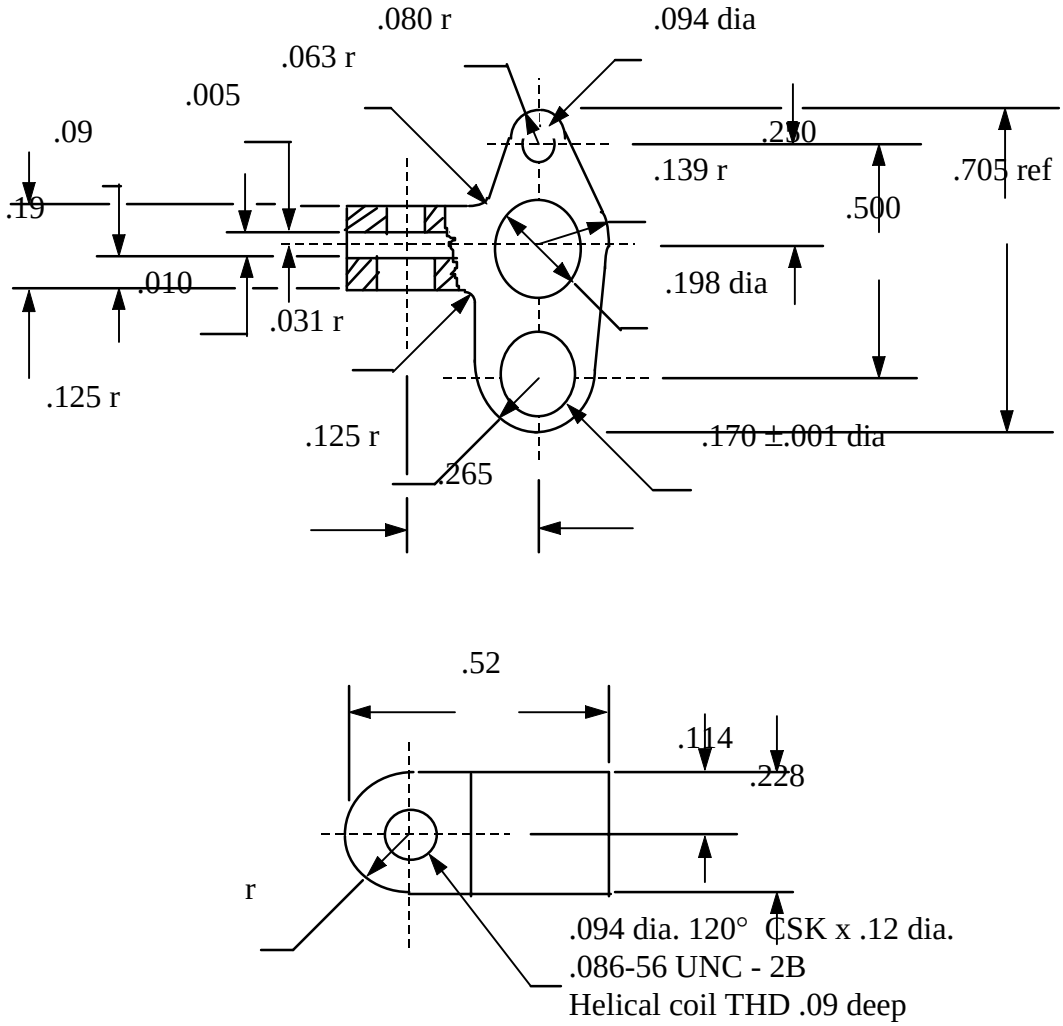


FIGURE 3. Aluminum Block (250 Joule mass).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49319A	**2. DOCUMENT DATE (YYMMDD)** 97/07/03

3. DOCUMENT TITLE DETECTOR-DEWAR, DT-591B/UA

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
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